Business Management Study Guide

Weeks 1 to 6 | Concepts and MCQs | Written in simple language with examples

Weeks 1 to 6

Concepts explained in simple language

This PDF is arranged week by week. Each week contains a clear Concepts section, a full MCQs section, and a separate Online Assessment Questions section for extra practice.

Week 1

Concepts

1. Foundational Concepts of BI&A

Business Intelligence and Analytics (BI&A) is the ability to convert raw data into useful intelligence that supports better business decisions. Instead of depending on guesswork, companies use BI&A to make choices based on evidence and strategy.

Informed decisions are made using data, analysis, and clear goals. Uninformed decisions depend on luck, intuition, or assumptions. The purpose of BI&A is to reduce uncertainty and move decisions toward the informed side.

User-generated data is data created by users while using digital platforms. Searches on Google, product views on Amazon, clicks on websites, and app activity are all examples. Many companies provide free services because the data created by users becomes highly valuable for analytics.

Analytics is the deeper engine inside BI&A. BI helps organize, store, and report data, while analytics uses algorithms and models to discover patterns, explain behavior, and predict future outcomes.

Data as a strategic asset means that data itself creates business value. In modern digital businesses, a company can be worth billions not only because of current revenue, but because of the user data, network connections, and future opportunities that data makes possible.

2. Patterns and the Data Revolution

Descriptive analytics answers the question, ?What is happening?? It summarizes historical data and gives a factual picture of business performance. For example, a monthly sales report is descriptive analytics.

Explanatory or diagnostic analytics asks, ?Why is it happening?? It goes beyond observation and tests possible reasons behind a pattern. For example, if sales drop in one region, the analyst studies pricing, competition, and customer behavior to explain the cause.

Predictive analytics asks, ?What is likely to happen next?? It uses past data and models to forecast future outcomes such as churn, sales, or fraud risk.

Prescriptive analytics asks, ?What should we do?? It recommends actions based on predicted outcomes. For example, it may suggest giving a retention offer to customers with high churn risk.

Cognitive analytics uses AI-like reasoning to simulate more human-like decision support, especially when handling complex or unstructured information.

A hypothesis is a testable explanation proposed from limited evidence. It helps analysts move from noticing a pattern to examining whether a real cause exists.

The Long Tail describes a business model in which many low-demand niche items together can generate as much value as a few best-selling items. This became practical because digital platforms make storage, search, and distribution much cheaper than in physical stores.

Macro-forces behind the data revolution are the falling costs of processing power, data storage, and data transmission. These three forces made it possible to work with massive amounts of data.

3. BI&A Architecture and Course Pillars

BI&A architecture is the technical setup that stores, moves, organizes, and analyzes data. It includes databases, data warehouses, ETL systems, and OLAP systems. It acts like the factory where raw data is prepared for analysis.

Business-to-analytics translation is the skill of converting a vague business problem into a clear data problem. For example, ?sales are down? must be translated into measurable questions such as which products, which customers, which regions, and during which time period are responsible.

Machine learning refers to methods that learn patterns from data and use them for prediction or classification. In this course context, it is closely related to data mining.

The course learning objectives move from understanding architecture, to translating business problems, to applying techniques, and finally to using practical software tools.

Software tools such as Python, Jupyter Notebook, R, RStudio, and SQL are used to work with data, run analyses, and apply models on large datasets.

4. Vocabulary of Analytics

ETL stands for Extract, Transform, Load. First, data is extracted from sources. Then it is transformed by cleaning, formatting, and standardizing it. Finally, it is loaded into a data warehouse or target system.

Data warehousing is the practice of storing cleaned, integrated, structured data in a central place for analysis. A warehouse is designed for historical queries and decision support rather than daily operations.

OLAP, or Online Analytical Processing, supports multidimensional analysis. It lets managers slice and dice data by different dimensions such as region, time, product, or customer segment.

Data mining, also called KDD or Knowledge Discovery in Databases, is the process of finding hidden patterns, relationships, and useful knowledge in large datasets.

Big Data is data characterized by high volume, velocity, and variety. Traditional tools struggle to handle it, so specialized systems are often required.

The three pillars of success in analytics are technology, science, and domain knowledge. Good tools alone are not enough. The analyst also needs statistical thinking and understanding of the business context.

> *Example: A company may have millions of customer clicks. BI organizes that information into dashboards, while analytics studies it to predict which customers are likely to buy, leave, or respond to an offer.*

MCQs

1. What does ETL stand for in BI&A?

A) Enter, Transfer, Load
B) Extract, Transform, Load
C) Execute, Train, Learn
D) Evaluate, Test, Log

Answer: B

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2. Which stage of ETL is mainly responsible for cleaning and standardizing data?

A) Extract
B) Transform
C) Load
D) Store

Answer: B

---

3. What is the main purpose of ETL in BI&A?

A) To build machine learning models
B) To prepare data for analysis
C) To create dashboards only
D) To store raw data without changes

Answer: B

---

4. For what purpose is R mainly used?

A) Web development
B) System programming
C) Statistical analysis
D) Hardware design

Answer: C

---

5. Which statement best describes R?

A) It is a database management system.
B) It is a programming language used for statistics.
C) It is an operating system.
D) It is a networking tool.

Answer: B

---

6. What does the Long Tail concept mainly focus on?

A) Selling only popular products
B) Selling many niche products
C) Reducing product variety
D) Increasing storage cost

Answer: B

---

7. Which factor mainly enables the Long Tail model?

A) Limited inventory
B) High storage costs
C) Internet and digital platforms
D) Fewer products

Answer: C

---

8. In the Long Tail model, where does a large share of revenue come from?

A) Only from top-selling products
B) Only from advertisements
C) From many low-demand products together
D) From government policies

Answer: C

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9. Which type of analytics answers the question ?What happened??

A) Predictive analytics
B) Prescriptive analytics
C) Descriptive analytics
D) Cognitive analytics

Answer: C

---

10. What is a hypothesis in analytics?

A) A proven fact
B) A random guess
C) A testable explanation
D) A software tool

Answer: C

---

11. Which ETL step collects data from source systems?

A) Transform
B) Extract
C) Load
D) Analyze

Answer: B

---

12. Where is data commonly stored after the ETL process is completed?

A) OLAP only
B) Data warehouse
C) Browser
D) API

Answer: B

---

13. What is a data warehouse?

A) Temporary storage for random files
B) Central storage designed for analysis
C) A raw data source only
D) A programming tool

Answer: B

---

14. For what purpose is OLAP mainly used?

A) Coding applications
B) Storing data permanently
C) Multidimensional analysis
D) Data collection

Answer: C

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15. The phrase ?slicing and dicing? is most closely associated with which technology?

A) ETL
B) OLAP
C) SQL
D) Python

Answer: B

---

16. Big Data is commonly defined by which three characteristics?

A) Speed only
B) Size only
C) Volume, velocity, and variety
D) Value only

Answer: C

---

17. Which option is not one of the common characteristics of Big Data?

A) Volume
B) Velocity
C) Variety
D) Visualization

Answer: D

---

18. What is predictive analytics mainly used for?

A) Explaining the past only
B) Describing the present only
C) Predicting future outcomes
D) Storing data

Answer: C

---

19. What does prescriptive analytics mainly help a business do?

A) Store data
B) Predict outcomes only
C) Suggest actions
D) Clean data

Answer: C

---

20. What does cognitive analytics involve?

A) Manual calculations only
B) AI-based decision support
C) Data storage only
D) Reporting only

Answer: B

---

21. What does the Pareto Principle usually state?

A) 50/50 rule
B) 70/30 rule
C) 80/20 rule
D) 90/10 rule

Answer: C

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22. In BI systems, dashboards mainly belong to which area?

A) Data collection
B) Data storage
C) Data visualization
D) Data mining

Answer: C

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23. What is the main purpose of data mining?

A) To store data
B) To find patterns
C) To delete data
D) To transfer data

Answer: B

---

24. What does KDD stand for?

A) Knowledge Data Design
B) Knowledge Discovery in Databases
C) Key Data Definition
D) Kernel Data Design

Answer: B

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25. What is SQL mainly used for?

A) Machine learning
B) Database querying
C) Web design
D) Networking

Answer: B

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26. Which tool is commonly used with R?

A) VS Code only
B) RStudio
C) Eclipse
D) IntelliJ

Answer: B

---

27. Which pillar is necessary for analytics success?

A) Luck
B) Domain knowledge
C) Hardware only
D) Marketing alone

Answer: B

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28. What is machine learning?

A) Data storage
B) Rule-based coding only
C) Learning patterns from data
D) Manual analysis only

Answer: C

---

29. What does business-to-analytics translation mean?

A) Writing code only
B) Converting business problems into data problems
C) Storing data in a warehouse
D) Designing a user interface

Answer: B

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30. What is user-generated data?

A) Data created by servers only
B) Data created by users through actions such as clicks and searches
C) Data created by hardware only
D) Data created by sensors only

Answer: B

---

31. Which option is an example of descriptive analytics?

A) Forecasting next year?s sales
B) Recommending the best action
C) A monthly sales report
D) An AI chatbot

Answer: C

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32. What question does diagnostic analytics answer?

A) What will happen?
B) Why did it happen?
C) What should be done?
D) Where is the data stored?

Answer: B

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33. Which type of analytics uses machine learning most often to estimate future outcomes?

A) Descriptive analytics
B) Diagnostic analytics
C) Predictive analytics
D) Data entry

Answer: C

---

34. What is the main role of business intelligence?

A) To build hardware
B) To organize and present data
C) To invent new ML algorithms
D) To manage networks

Answer: B

---

35. Which option is an example of user-generated data?

A) Server logs only
B) Product manufacturing data
C) User search queries
D) CPU usage

Answer: C

---

36. Why is data called a strategic asset?

A) Because it is easy to delete
B) Because it generates insight and value
C) Because it reduces storage needs
D) Because it replaces hardware

Answer: B

---

37. Which macro-force is strongly associated with Moore?s Law?

A) Storage only
B) Processing power
C) Internet only
D) Software updates

Answer: B

---

38. Which option is not part of ETL?

A) Extract
B) Transform
C) Load
D) Learn

Answer: D

---

39. In which ETL stage does data cleaning mainly happen?

A) Extract
B) Transform
C) Load
D) Analyze

Answer: B

---

40. Which system mainly handles daily transactions?

A) Data warehouse
B) OLAP
C) Operational database
D) ETL

Answer: C

---

41. A data warehouse is optimized for which purpose?

A) Fast transactions
B) Historical analysis
C) Gaming
D) Networking

Answer: B

---

42. What does OLAP allow a user to do?

A) Write application code
B) Analyze data from multiple dimensions
C) Store raw data only
D) Clean data

Answer: B

---

43. Which option is an example of prescriptive analytics?

A) A sales report
B) A root cause analysis
C) A suggested discount strategy
D) Data storage

Answer: C

---

44. Which option is a demand-side factor supporting the Long Tail?

A) Warehouses
B) Inventory racks
C) Recommendation systems
D) Manufacturing

Answer: C

---

45. Which option is a supply-side factor supporting the Long Tail?

A) Search engines
B) Recommender systems
C) Unlimited digital inventory
D) User behavior only

Answer: C

---

46. What does Big Data usually require?

A) Traditional tools only
B) Specialized processing systems
C) Manual entry only
D) Paper storage

Answer: B

---

47. Which language is widely used for analytics work?

A) HTML
B) Python
C) CSS
D) Assembly

Answer: B

---

48. In which area is R especially strong?

A) Networking
B) Statistics
C) Hardware
D) Gaming

Answer: B

---

49. Which activity is an important part of the analytics process?

A) Guessing without evidence
B) Hypothesis testing
C) Ignoring data
D) Random decision making

Answer: B

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50. What is the main goal of analytics?

A) To store data
B) To delete data
C) To extract insights
D) To increase hardware

Answer: C

Online Assessment Questions

1. What does AI stand for?

A) Automated Intelligence
B) Artificial Intelligence
C) Advanced Integration
D) Algorithmic Interface

Answer: B) Artificial Intelligence

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2. Which type of AI performs a specific task only?

A) AGI
B) ASI
C) ANI
D) Sentient AI

Answer: C) ANI

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3. What does supervised learning require?

A) No data
B) Labeled data
C) Random guessing
D) Only images

Answer: B) Labeled data

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4. What is unsupervised learning used to do?

A) Predict exact outputs
B) Label data manually
C) Find patterns in data
D) Train robots

Answer: C) Find patterns in data

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5. On what is reinforcement learning based?

A) Rewards and penalties
B) Fixed rules
C) Human instructions only
D) No feedback

Answer: A) Rewards and penalties

---

6. What is overfitting in AI?

A) A model performs well on all data
B) A model memorizes training data but fails on new data
C) A model ignores data
D) A model is too simple

Answer: B) A model memorizes training data but fails on new data

---

7. Which of the following is an example of AI in daily life?

A) Calculator
B) Voice assistants like Siri
C) Light bulb
D) Keyboard

Answer: B) Voice assistants like Siri

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8. What is a dataset?

A) A single number
B) A collection of data
C) A computer program
D) A hardware device

Answer: B) A collection of data

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9. What does bias in AI mean?

A) Perfect accuracy
B) Random output
C) Unfair or skewed results
D) Faster processing

Answer: C) Unfair or skewed results

---

10. What is the goal of machine learning?

A) Replace humans completely
B) Learn patterns from data to make decisions
C) Store large files
D) Build hardware

Answer: B) Learn patterns from data to make decisions

Week 2

Concepts

1. Bizocity Case and Analytics

Customer acquisition means getting new customers. In the Bizocity case, older methods such as random calling wasted time and money because many contacted numbers were not useful leads.

As-Is describes the current or old system. To-Be describes the improved future system. In this case, the old method depended on manual calling, while the better method used data to target likely business customers.

Call Detail Data (CDD) contains metadata about calls, such as the numbers involved, time of call, and duration. It does not include the actual voice content of the call.

Bizocity score is a score between 0 and 1 that indicates how likely a phone number belongs to a business. A higher score means the number behaves more like a business number.

Behavior indicators are clues from usage patterns. Business numbers often show daytime calling, longer calls, and interactions with other business-like numbers.

RFM analysis studies behavior using Recency, Frequency, and Monetary value. In this context, recency means how recently the number was seen, frequency means how often it appeared, and monetary value is represented by call duration.

The key idea is that we do not always need to ask directly whether a customer is a good target. Sometimes behavior itself gives a strong signal.

2. Business Intelligence Architecture

BI architecture is the full system that takes raw data, prepares it, analyzes it, and turns it into decision support.

OLTP stands for Online Transaction Processing. These systems handle daily operations such as orders, payments, and bookings. They are designed for fast and frequent transaction processing.

ETL prepares data by extracting it from sources, transforming it into clean and consistent form, and loading it into a storage system designed for analysis.

A data warehouse stores integrated, historical, structured data for analysis.

OLAP helps analyze the stored data from different dimensions.

Data mining finds patterns and makes predictions.

A data mart is a smaller part of a warehouse created for one department or business function, such as marketing or finance.

A data lake stores raw data, including unstructured data such as social media posts, video, audio, and logs.

Inmon?s approach is top-down: build the enterprise warehouse first, then create smaller marts. Kimball?s approach is bottom-up: build smaller data marts first and integrate them later.

3. Data Management and RDBMS

Data management means handling data carefully so that it remains correct, complete, useful, and consistent.

RDBMS, or Relational Database Management System, stores data in tables linked by relationships.

An entity is a real-world object represented in data, such as a customer, product, or order.

An attribute is a property of an entity, such as customer name, product price, or order date.

A primary key uniquely identifies each row in a table and cannot be duplicated or null.

A foreign key connects one table to another by referencing a key in the related table.

Entity integrity ensures that each record is uniquely identifiable. Referential integrity ensures that relationships remain valid. Data integrity means that data remains accurate and consistent overall.

4. OLTP and Database Design

Cardinality describes the type of relationship between entities, such as one-to-one, one-to-many, or many-to-many.

In a one-to-many relationship, the foreign key is placed on the many side. For example, one customer can place many orders, so the customer key appears in the order table.

A many-to-many relationship cannot usually be implemented directly. It needs an associative entity or junction table to connect the two sides.

Normalization removes repetition and improves consistency.

ACID properties make transactions reliable: Atomicity means all or nothing, Consistency means valid state, Isolation means transactions do not disturb one another, and Durability means committed changes are permanently saved.

OLTP vs data warehouse: OLTP is for current daily work; the warehouse is for historical analysis.

> *Example: A telecom company can examine daytime calling frequency and call duration to decide whether a number is likely to belong to a business, without ever listening to the call itself.*

MCQs

1. What does customer acquisition mean?

A) Deleting customers
B) Getting new customers
C) Storing data
D) Marketing only

Answer: B

---

2. What does the term ?As-Is? refer to?

A) The future system
B) The ideal system
C) The current system
D) The final output

Answer: C

---

3. What does the term ?To-Be? refer to?

A) The current system
B) The improved future system
C) The old method
D) The raw data

Answer: B

---

4. What does Call Detail Data store?

A) The voice content of calls
B) Call metadata
C) Messages only
D) Emails only

Answer: B

---

5. What is the range of a Bizocity score?

A) 1 to 100
B) 0 to 10
C) 0 to 1
D) -1 to 1

Answer: C

---

6. What does a high Bizocity score indicate?

A) The number is personal.
B) The number is likely to be a business number.
C) The number is invalid.
D) The number is unknown.

Answer: B

---

7. During which period do business calls usually happen most often?

A) At night
B) In the early morning only
C) During office hours
D) At random times only

Answer: C

---

8. In RFM analysis, what does Recency mean?

A) Total money
B) Last seen time
C) Frequency
D) Duration only

Answer: B

---

9. In RFM analysis, what does Frequency mean?

A) Value
B) Time
C) How often something happens
D) Cost

Answer: C

---

10. In the Bizocity case, what represents Monetary value in RFM analysis?

A) Salary
B) Call duration
C) Number of calls only
D) Profit

Answer: B

---

11. What does BI architecture mainly do?

A) It deletes data.
B) It converts data into insights.
C) It only stores data.
D) It only encrypts data.

Answer: B

---

12. What is OLTP used for?

A) Analysis
B) Daily transactions
C) Prediction
D) Long-term archival only

Answer: B

---

13. Which statement best describes OLTP systems?

A) They are slow.
B) They are fast for operational work.
C) They are mainly historical.
D) They store only raw data.

Answer: B

---

14. What does ETL stand for?

A) Extract Transfer Load
B) Extract Transform Load
C) Enter Transform Load
D) Extract Test Load

Answer: B

---

15. What happens during the Transform stage of ETL?

A) Data is deleted.
B) Data is cleaned and formatted.
C) Data is stored permanently.
D) Data is analyzed.

Answer: B

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16. What does a data warehouse mainly store?

A) Raw unprocessed data only
B) Clean structured data
C) Images only
D) Temporary data only

Answer: B

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17. A data warehouse is mainly used for which purpose?

A) Transactions
B) Analysis
C) Calling customers
D) Messaging

Answer: B

---

18. What is OLAP mainly used for?

A) Transactions
B) Data entry
C) Analysis
D) Storage only

Answer: C

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19. What is data mining mainly used for?

A) Storage
B) Pattern finding and prediction
C) Transactions
D) Cleaning only

Answer: B

---

20. What is a data mart?

A) The complete warehouse
B) A subset of data for a specific function
C) Raw storage only
D) An OLTP system

Answer: B

---

21. What does a data lake mainly store?

A) Only structured data
B) Only tables
C) Raw data
D) Only clean data

Answer: C

---

22. Which approach is associated with Inmon?

A) Bottom-up
B) Top-down
C) No ETL
D) Data lake only

Answer: B

---

23. Which approach is associated with Kimball?

A) Top-down
B) Bottom-up
C) No warehouse
D) No ETL

Answer: B

---

24. In an RDBMS, how is data mainly stored?

A) In files only
B) In tables
C) In images
D) In audio records

Answer: B

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25. What is an entity in database design?

A) A property only
B) A table command
C) A real-world object
D) Data storage space

Answer: C

---

26. What is an attribute in database design?

A) An object
B) A detail or property
C) A table type
D) A system process

Answer: B

---

27. What is a primary key?

A) A duplicate value
B) A unique identifier
C) A foreign link
D) A table name

Answer: B

---

28. Which statement is true about a primary key?

A) It may be null.
B) It cannot be null.
C) It is always duplicated.
D) It is optional in every table.

Answer: B

---

29. What is a foreign key?

A) A unique ID for the same table
B) A link between tables
C) A table itself
D) A type of raw data

Answer: B

---

30. What does entity integrity ensure?

A) Linking only
B) Uniqueness of records
C) Speed of queries
D) Larger storage

Answer: B

---

31. What does referential integrity ensure?

A) Unique data only
B) Valid links between tables
C) Fast queries only
D) More storage

Answer: B

---

32. What does data integrity mean?

A) Speed
B) Accuracy and consistency
C) Size only
D) Format only

Answer: B

---

33. What does cardinality define in database design?

A) Data size
B) Relationship type
C) Speed
D) Storage type

Answer: B

---

34. What does a 1:N relationship mean?

A) One to many
B) Many to one only
C) One to one
D) None of these

Answer: A

---

35. In a 1:N relationship, where is the foreign key placed?

A) On the one side
B) On the many side
C) On both sides
D) In a new table always

Answer: B

---

36. What does an N:N relationship mean?

A) One to one
B) One to many
C) Many to many
D) No relationship

Answer: C

---

37. What is needed to implement an N:N relationship properly?

A) A primary key only
B) A foreign key only
C) A new junction table
D) Deleted data

Answer: C

---

38. What is a junction table?

A) A primary table
B) A linking table
C) A raw table
D) A data lake

Answer: B

---

39. What does normalization mainly do?

A) It increases data.
B) It reduces repetition.
C) It deletes all data.
D) It encrypts data.

Answer: B

---

40. In ACID properties, what does Atomicity mean?

A) Save forever
B) All or nothing
C) Isolation
D) Accuracy only

Answer: B

---

41. In ACID properties, what does Consistency mean?

A) A valid state is maintained.
B) Data is saved forever.
C) Transactions are isolated only.
D) Data is deleted.

Answer: A

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42. In ACID properties, what does Isolation mean?

A) Data is saved permanently.
B) Transactions remain separate from each other.
C) Data is deleted.
D) Data is cleaned.

Answer: B

---

43. In ACID properties, what does Durability mean?

A) Data is temporary.
B) Committed changes are permanently saved.
C) Data is deleted.
D) Queries become faster.

Answer: B

---

44. What does OLTP mainly handle?

A) Analysis
B) Daily work
C) Prediction
D) Storage only

Answer: B

---

45. What does a data warehouse mainly handle?

A) Daily work
B) Analysis
C) Calls only
D) Messages only

Answer: B

---

46. OLTP data is usually which type of data?

A) Historical
B) Current
C) Raw only
D) Clean only

Answer: B

---

47. Data warehouse data is usually which type of data?

A) Current only
B) Historical
C) Temporary
D) Raw only

Answer: B

---

48. What is the main goal of BI?

A) To store data only
B) To support decision making
C) To delete data
D) To encrypt everything

Answer: B

---

49. What does ETL make data ready for?

A) Raw use only
B) Clean use and analysis
C) Deletion
D) Random storage

Answer: B

---

50. How does the Bizocity score help a company?

A) By improving storage only
B) By helping target customers
C) By deleting data
D) By creating backups only

Answer: B

Online Assessment Questions

1. What is data leakage in machine learning?

A) Using too much data
B) Training with future or test data information
C) Reducing dataset size
D) Removing features

Answer: B) Training with future or test data information

---

2. When does underfitting occur?

A) When the model is too complex
B) When the model learns noise
C) When the model is too simple to capture patterns
D) When the model overtrains

Answer: C) When the model is too simple to capture patterns

---

3. What does a hidden layer in an ANN do?

A) It stores output.
B) It deletes data.
C) It extracts intermediate features.
D) It only normalizes data.

Answer: C) It extracts intermediate features

---

4. What is the purpose of activation functions?

A) Increase data size
B) Introduce non-linearity
C) Store weights
D) Reduce layers

Answer: B) Introduce non-linearity

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5. What is gradient descent used for?

A) Increasing error
B) Updating weights to minimize error
C) Adding noise
D) Removing data

Answer: B) Updating weights to minimize error

---

6. Which loss function is commonly used for regression?

A) Cross-entropy
B) Squared error
C) Hinge loss
D) Log loss

Answer: B) Squared error

---

7. What may increasing model complexity lead to?

A) Underfitting
B) Overfitting
C) No change
D) Data loss

Answer: B) Overfitting

---

8. What is the role of bias in a neuron?

A) Removes input
B) Adds flexibility to the model
C) Deletes weights
D) Normalizes output

Answer: B) Adds flexibility to the model

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9. What does time series data include?

A) Random unordered data
B) Data with time dependency
C) Only images
D) Only text

Answer: B) Data with time dependency

---

10. Why are neural networks called ?black box? models?

A) They are physically black
B) They are simple
C) Their internal workings are hard to interpret
D) They use no data

Answer: C) Their internal workings are hard to interpret

Week 3

Concepts

1. DBMS, RDBMS, and SQL Basics

DBMS is a system that stores and manages data. RDBMS is a relational DBMS that stores data in tables and connects those tables using relationships.

SQL is the language used to communicate with relational databases. It can be used to retrieve, add, update, and structure data. SQL commands are generally not case-sensitive, so select and SELECT mean the same thing.

In a table, a row is one full record, while a column is one field or attribute.

2. ACID and Keys

ACID properties make transactions reliable. Atomicity means a transaction happens completely or not at all. Consistency means the database remains valid. Isolation means concurrent transactions do not interfere with each other. Durability means once a transaction is committed, it stays saved even after a failure.

A super key is any set of attributes that uniquely identifies a row. A candidate key is a minimal super key. A primary key is the chosen candidate key used as the main identifier. A foreign key links one table to another.

A composite key uses multiple columns together to identify a row. A compound key is a type of composite key that includes at least one foreign key.

3. SQL Commands and JOINs

DDL or Data Definition Language is used for structure, such as CREATE and ALTER.

DML or Data Manipulation Language is used to insert and modify data, such as INSERT and UPDATE.

DQL or Data Query Language is used to retrieve data, mainly through SELECT.

INNER JOIN returns matching rows from both tables. LEFT JOIN returns all rows from the left table and matching rows from the right. RIGHT JOIN returns all rows from the right table and matching rows from the left. FULL JOIN returns all rows from both sides.

4. Normalization

Normalization is the process of organizing data to reduce redundancy and avoid anomalies.

Insertion anomaly happens when you cannot add data properly without adding unrelated data. Deletion anomaly happens when deleting one fact removes another useful fact. Update anomaly happens when the same value is stored in many places and is not updated consistently.

First Normal Form (1NF) requires each cell to contain only one value.

Second Normal Form (2NF) removes partial dependency, meaning a non-key attribute should depend on the full key, not part of it.

Third Normal Form (3NF) removes transitive dependency, meaning non-key attributes should not depend on other non-key attributes.

BCNF is a stronger version of 3NF in which the determinant on the left side of every dependency must be a super key.

The trade-off is that higher normalization improves cleanliness and integrity, but queries may become slower because more joins are needed.

5. OLAP Concepts

OLTP handles transactional work. OLAP is designed for analysis and reporting.

Metadata is data about data, such as table names, column names, and data types.

A fact table stores measurable values like sales amount, quantity, or profit. A dimension table stores descriptive context like time, product, or location.

A star schema is denormalized and has fewer joins, so it is often faster for analysis. A snowflake schema is more normalized and uses more joins.

A data cube stores data in multiple dimensions. Common operations are slice for one fixed value, dice for a range or subset, drill-down for more detail, roll-up for summary, and pivot for changing the view.

Materialization means precomputing and storing results in advance so queries can run faster later.

Example: In a sales cube, you may analyze sales by time, product, and city. Viewing only one city is slicing, while looking at selected cities and products together is dicing.

MCQs

1. Which system stores data in tables with relationships?

A) DBMS
B) RDBMS
C) Operating system
D) API

Answer: B

---

2. What is SQL used for?

A) Designing a user interface
B) Managing databases
C) Compiling code
D) Networking

Answer: B

---

3. Which statement is correct about SQL commands?

A) SQL is case-sensitive.
B) SQL is generally case-insensitive.
C) SQL must be written only in uppercase.
D) SQL must be written only in lowercase.

Answer: B

---

4. What is another name for a row in a table?

A) Field
B) Column
C) Record
D) Table

Answer: C

---

5. What is another name for a column in a table?

A) Record
B) Field
C) Row
D) Key

Answer: B

---

6. Which ACID property ensures ?all or nothing? behavior?

A) Consistency
B) Atomicity
C) Isolation
D) Durability

Answer: B

---

7. Which ACID property ensures that data remains saved after a crash?

A) Atomicity
B) Isolation
C) Durability
D) Consistency

Answer: C

---

8. What is a minimal super key called?

A) Primary key
B) Candidate key
C) Foreign key
D) Composite key

Answer: B

---

9. Which statement is true about a primary key?

A) It can be null.
B) It can be duplicated.
C) It must be unique and not null.
D) It is optional.

Answer: C

---

10. What is a foreign key used for?

A) Deleting data
B) Linking tables
C) Sorting data
D) Printing data

Answer: B

---

11. What is DDL mainly used for?

A) Data manipulation
B) Structure definition
C) Querying only
D) Security only

Answer: B

---

12. To which language category does INSERT belong?

A) DDL
B) DML
C) DQL
D) TCL

Answer: B

---

13. To which language category does SELECT belong?

A) DML
B) DQL
C) DDL
D) TCL

Answer: B

---

14. What does an INNER JOIN return?

A) All records
B) Matching records
C) Only left table records
D) Only right table records

Answer: B

---

15. What does a LEFT JOIN return?

A) Only matching records
B) All rows from the left table and matching rows from the right
C) All rows from the right table only
D) No rows

Answer: B

---

16. What does normalization mainly reduce?

A) Speed only
B) Redundancy
C) Tables only
D) Security risks only

Answer: B

---

17. A multi-valued attribute violates which normal form?

A) 1NF
B) 2NF
C) 3NF
D) BCNF

Answer: A

---

18. Which dependency is removed by Second Normal Form?

A) Redundancy only
B) Partial dependency
C) Transitive dependency
D) Keys

Answer: B

---

19. Which dependency is removed by Third Normal Form?

A) Partial dependency
B) Transitive dependency
C) Atomicity issues
D) Key constraints

Answer: B

---

20. BCNF requires the determinant to be which type of key?

A) Candidate key only
B) Super key
C) Foreign key
D) Primary key only

Answer: B

---

21. What can deletion anomaly cause?

A) Duplicate data
B) Data loss
C) Slow queries only
D) Error logs only

Answer: B

---

22. What is OLTP mainly used for?

A) Analysis
B) Transactions
C) Backup only
D) Reporting only

Answer: B

---

23. What is OLAP mainly used for?

A) Daily operations
B) Analysis
C) Deletion
D) Coding

Answer: B

---

24. What does metadata mean?

A) Raw data
B) Data about data
C) Tables only
D) Queries only

Answer: B

---

25. What does a fact table mainly store?

A) Text descriptions
B) Numeric measures
C) Keys only
D) Queries

Answer: B

---

26. What does a dimension table mainly store?

A) Measures only
B) Context and descriptions
C) Keys only
D) Queries

Answer: B

---

27. Which statement best describes a star schema?

A) It is highly normalized.
B) It is denormalized.
C) It is deleted.
D) It is random.

Answer: B

---

28. Which statement best describes a snowflake schema?

A) It is denormalized.
B) It is normalized.
C) It is flat only.
D) It is random.

Answer: B

---

29. What is a data cube?

A) One-dimensional data
B) Two-dimensional data only
C) Multi-dimensional data
D) A flat file

Answer: C

---

30. What does the slice operation do in OLAP?

A) It selects a range of values.
B) It fixes one value of a dimension.
C) It rotates the view.
D) It adds more detail.

Answer: B

---

31. What does the dice operation do in OLAP?

A) It selects one value only.
B) It selects a range or subset.
C) It rotates the view.
D) It adds detail only.

Answer: B

---

32. What does drill-down mean?

A) Summary
B) More detail
C) Delete data
D) Rotate the view

Answer: B

---

33. What does roll-up mean?

A) Detail
B) Summary
C) Delete data
D) Rotate the view

Answer: B

---

34. What does pivot mean in OLAP?

A) Delete data
B) Rotate the view
C) Insert rows
D) Update columns

Answer: B

---

35. What does materialization mean?

A) Deleting data
B) Pre-computing results
C) Normalizing tables
D) Storing logs only

Answer: B

---

36. What does a composite key use?

A) One column
B) Multiple columns
C) No columns
D) Only foreign keys

Answer: B

---

37. What is special about a compound key?

A) It uses only a primary key.
B) It includes a foreign key as part of the key.
C) It uses only a candidate key.
D) It contains no relationship.

Answer: B

---

38. What is GROUP BY used for in SQL?

A) Sorting
B) Filtering
C) Aggregation
D) Joining

Answer: C

---

39. What is ORDER BY used for in SQL?

A) Sorting
B) Filtering
C) Grouping
D) Joining

Answer: A

---

40. What does the AVG() function calculate?

A) Sum
B) Count
C) Average
D) Maximum

Answer: C

---

41. What does the SUM() function calculate?

A) Average
B) Total
C) Count
D) Minimum

Answer: B

---

42. What does the COUNT() function calculate?

A) Total amount
B) Number of rows
C) Average
D) Maximum

Answer: B

---

43. From which type of key is the primary key usually chosen?

A) Super key
B) Candidate key
C) Foreign key
D) Composite key only

Answer: B

---

44. What is a super key?

A) A minimal identifier only
B) A unique identifier
C) A foreign key
D) An optional field

Answer: B

---

45. What does insertion anomaly mean?

A) Data cannot be inserted properly without problems.
B) Duplicate data appears.
C) Existing data gets deleted.
D) A query fails.

Answer: A

---

46. What does update anomaly mean?

A) Updates become inconsistent.
B) Data is lost.
C) Inserts always fail.
D) Queries stop running.

Answer: A

---

47. Which practice improves data integrity?

A) Redundancy
B) Normalization
C) Duplication
D) Deletion

Answer: B

---

48. Which approach can cause slower queries because of more joins?

A) Denormalization
B) Normalization
C) Indexing
D) Caching

Answer: B

---

49. Which schema usually has fewer joins?

A) Snowflake
B) Star
C) BCNF
D) 3NF

Answer: B

---

50. Which system is generally faster for analytics queries?

A) OLTP
B) OLAP
C) DBMS only
D) API

Answer: B

Online Assessment Questions

1. What is Porter?s Cost Leadership strategy focused on?

A) Creating unique products
B) Selling at lowest cost
C) Targeting niche markets
D) Increasing marketing

Answer: B) Selling at lowest cost

---

2. In the IT Strategic Grid, what does the ?Factory? quadrant mean?

A) High strategic impact and low operational impact
B) Low strategic impact and high operational impact
C) High impact on both
D) Low impact on both

Answer: B) Low strategic impact and high operational impact

---

3. In Porter?s Value Chain, how is IT classified?

A) As a primary activity
B) As a support activity
C) As an external activity
D) As a strategic layer only

Answer: B) As a support activity

---

4. What is a business model?

A) A marketing plan
B) How a firm creates and earns value
C) Only a financial report
D) An HR strategy

Answer: B) How a firm creates and earns value

---

5. Which of the following is a primary activity in the value chain?

A) HR Management
B) Technology Development
C) Inbound Logistics
D) Firm Infrastructure

Answer: C) Inbound Logistics

---

6. What does network effect mean?

A) A decrease in users reduces value
B) Value increases with more users
C) Value stays fixed
D) There is no relation with users

Answer: B) Value increases with more users

---

7. Who controls the rules in a digital platform?

A) Users
B) Government
C) Platform owner
D) Developers only

Answer: C) Platform owner

---

8. What does AI data monetization focus on?

A) Reducing data
B) Using data to generate value
C) Deleting data
D) Only storing data

Answer: B) Using data to generate value

---

9. Why do platforms add social features such as LinkedIn posts?

A) To reduce users
B) To increase engagement
C) To remove content
D) To lower costs

Answer: B) To increase engagement

---

10. What does frictionless entry in platforms help with?

A) Slowing growth
B) Rapid user growth
C) Reducing users
D) Increasing costs

Answer: B) Rapid user growth

Week 4

Concepts

1. Customer Churn and Survival Analysis

Customer churn happens when a customer stops using a product or service. If someone cancels a subscription or becomes inactive, that customer has churned.

Survival analysis studies the time until an event occurs. In marketing and customer management, the event is often churn. Instead of looking only at who left, survival analysis also looks at when they left.

Customer tenure is the length of time a customer stays with the company. Higher tenure usually means greater long-term value.

The half-life period is the time by which 50% of customers have churned. A short half-life suggests weak retention.

2. Why Retention Matters

Companies spend money to acquire customers. If customers leave quickly, the firm may not recover that cost. This is why retention is often more profitable than constantly acquiring new customers.

A small reduction in churn can lead to a large increase in profit because loyal customers tend to buy more over time, cost less to serve, and may bring referrals.

3. Tenure Timeline, Hazard Rate, and Survival Rate

To compare customers fairly, analysts often use a tenure timeline. Instead of using calendar dates like January or February, they use Month 0, Month 1, Month 2, and so on, starting from the customer?s join date.

Hazard rate H(t) is the probability that a customer leaves during a specific period, given that the customer has survived up to that period. It tells us the risk of churn ?now.?

Survival rate S(t) is the probability that a customer is still active at time t. Survival usually starts at 1 and decreases or stays the same over time.

4. CLV and Customer Equity

Customer Lifetime Value (CLV) is the total expected future profit from a customer.

A discount rate is applied because money expected in the future is worth less than money received today.

Customer equity is the total value of all customers together, which is the sum of their CLVs.

CLV is a forward-looking measure. It helps businesses decide how much to spend on acquisition, retention, and customer service.

5. Key Business Insight

The biggest profitability improvement often comes from increasing retention. Keeping customers longer increases tenure, increases CLV, and raises overall customer equity.

Example: If a streaming customer stays for 24 months instead of 6 months, the company earns more subscription revenue and has more time to recover acquisition cost and generate profit.

MCQs

1. What is customer churn?

A) Gaining customers
B) Losing customers
C) Advertising
D) Selling products

Answer: B

---

2. What does survival analysis study?

A) Profit only
B) Time until an event occurs
C) Revenue only
D) Pricing

Answer: B

---

3. What does customer tenure mean?

A) Age
B) Spending only
C) Time stayed
D) Profit only

Answer: C

---

4. What does the half-life period mean in customer analysis?

A) 25% of customers leave
B) 50% of customers leave
C) 75% of customers leave
D) 100% of customers leave

Answer: B

---

5. What is a likely result of high churn?

A) High profit
B) No change
C) Loss
D) Growth only

Answer: C

---

6. Survival analysis originally came from which field?

A) Physics
B) Medicine
C) Marketing
D) Artificial intelligence

Answer: B

---

7. On a tenure timeline, which month represents the starting point?

A) Month 1
B) Month 0
C) Month 10
D) Year 1

Answer: B

---

8. What does the hazard rate show?

A) Total customers
B) Risk of leaving
C) Profit only
D) Cost only

Answer: B

---

9. Which statement best describes the survival rate?

A) It is an instant random number.
B) It is always fixed.
C) It is a cumulative chance of still staying.
D) It is unrelated to tenure.

Answer: C

---

10. In survival analysis, what does population at risk mean?

A) Customers who already churned
B) Customers who are still active
C) New customers only
D) Old customers only

Answer: B

---

11. What does CLV stand for?

A) Customer Loss Value
B) Customer Lifetime Value
C) Cost Level Value
D) Customer Level Value

Answer: B

---

12. What is customer equity?

A) Profit only
B) The sum of all CLVs
C) Revenue only
D) Cost only

Answer: B

---

13. What does the discount rate account for?

A) Profit only
B) The time value of money
C) Cost only
D) Risk only

Answer: B

---

14. What is acquisition cost?

A) The cost to retain a customer
B) The cost to get a customer
C) The cost to sell a product only
D) The cost to produce a product

Answer: B

---

15. CLV mainly focuses on which time period?

A) Past only
B) Present only
C) Future
D) None

Answer: C

---

16. What is one of the best ways to increase CLV?

A) Advertising only
B) Retention
C) Price cuts only
D) Hiring more employees

Answer: B

---

17. What does high tenure usually imply?

A) Low profit
B) High profit potential
C) No change
D) Loss

Answer: B

---

18. In retail settings, what may define the stop time?

A) Contract end only
B) Inactivity
C) Payment date only
D) Signup date

Answer: B

---

19. What does a spike in hazard rate indicate?

A) Stability
B) Increased churn risk
C) Profit growth
D) Customer acquisition

Answer: B

---

20. The survival formula is typically based on which operation across periods?

A) Addition
B) Multiplication
C) Division
D) Subtraction

Answer: B

---

21. In basic CLV calculation, what cost is often subtracted?

A) Revenue
B) Production cost only
C) Acquisition cost
D) Profit

Answer: C

---

22. What is contribution margin?

A) Revenue only
B) Profit after direct cost
C) Cost only
D) Loss only

Answer: B

---

23. Why is direct marketing especially useful for CLV analysis?

A) Because it is always cheap
B) Because data is available
C) Because it is always fast
D) Because it is easy

Answer: B

---

24. What does the 5/25 rule suggest?

A) 5% loss always causes 25% loss
B) A 5% reduction in churn can create a large profit increase
C) 25% churn is normal
D) None of these

Answer: B

---

25. Which statement is true about the survival rate over time?

A) It always increases.
B) It decreases or stays the same.
C) It behaves randomly only.
D) It always becomes negative.

Answer: B

---

26. Churn is the opposite of which concept?

A) Profit
B) Retention
C) Cost
D) Growth

Answer: B

---

27. What does retention mean?

A) Losing customers
B) Keeping customers
C) Selling products only
D) Marketing only

Answer: B

---

28. Compared with present money, future money is generally considered to be:

A) Equal in value
B) Higher in value
C) Lower in value
D) Undefined

Answer: C

---

29. In CLV thinking, customers are treated as what?

A) A cost only
B) An asset
C) A loss only
D) An expense only

Answer: B

---

30. In many businesses, Year 1 CLV is often what?

A) Positive
B) Negative
C) Zero always
D) Infinite

Answer: B

---

31. Hazard rate depends mainly on what?

A) Time
B) Cost only
C) Price only
D) Advertising only

Answer: A

---

32. What does a survival curve show?

A) Profit
B) Customers staying over time
C) Cost
D) Advertising spend

Answer: B

---

33. What is the effect of higher retention on CLV?

A) Lower CLV
B) Higher CLV
C) No change
D) Immediate loss

Answer: B

---

34. What does early churn usually mean for the company?

A) Profit
B) Loss
C) Growth
D) Stability

Answer: B

---

35. What effect does discounting have?

A) It reduces present value.
B) It reduces future value when converted to present terms.
C) It increases cost only.
D) It increases profit only.

Answer: B

---

36. Customer equity is closely linked to what?

A) HR decisions
B) Firm value
C) Salary structure
D) Manufacturing cost

Answer: B

---

37. Which statement best describes hazard rate?

A) It is total lifetime risk.
B) It is period-specific churn risk.
C) It means no risk.
D) It is always fixed.

Answer: B

---

38. What is the survival rate at the start of the tenure timeline?

A) 0
B) 1
C) 0.5
D) Random

Answer: B

---

39. Which type of profit does CLV mainly include?

A) Past profit only
B) Future profit
C) Cost only
D) None

Answer: B

---

40. Better retention usually reduces what?

A) Profit
B) Cost of replacing lost customers
C) Revenue
D) Growth

Answer: B

---

41. Long-term customers often do which of the following?

A) Cost more only
B) Buy more
C) Leave earlier
D) None

Answer: B

---

42. Referrals are more likely to come from which customers?

A) New users only
B) Loyal users
C) Ads only
D) Discounts only

Answer: B

---

43. What does a high hazard rate mean?

A) Safe customer base
B) Risky customer base
C) Higher profit automatically
D) Higher growth automatically

Answer: B

---

44. What does a low survival rate indicate?

A) Strong loyalty
B) High churn
C) High profit
D) Growth

Answer: B

---

45. For what is CLV especially useful?

A) HR planning
B) Marketing decisions
C) Coding
D) UI design

Answer: B

---

46. What is the event of a customer leaving usually called?

A) Entry
B) Exit
C) Churn
D) Join

Answer: C

---

47. If the discount rate increases, what usually happens to CLV?

A) It increases.
B) It decreases.
C) It stays the same.
D) It becomes zero.

Answer: B

---

48. What is a likely result of better retention?

A) Less revenue
B) More revenue
C) No change
D) Loss

Answer: B

---

49. Which group is excluded from the population at risk?

A) Active users
B) Customers who already churned
C) New users
D) Old users

Answer: B

---

50. What is the main goal of CLV analysis?

A) Past analysis only
B) Predicting future customer value
C) Cost control only
D) Hiring decisions

Answer: B

Online Assessment Questions

1. What is an ecosystem business model?

A) One company controls everything
B) Direct buyer-seller only
C) Multiple stakeholders create value together
D) Only a data-selling platform

Answer: C) Multiple stakeholders create value together

2. What does content-based filtering recommend on the basis of?

A) Other users? behavior
B) The user?s own past activity
C) Random selection
D) Demographics only

Answer: B) The user?s own past activity

3. Why are GPUs important in AI?

A) They store data.
B) They enable faster parallel computation.
C) They reduce data.
D) They replace software.

Answer: B) Faster parallel computations

4. What is ImageNet used for?

A) Text processing
B) A large-scale image dataset
C) Audio storage
D) Networking

Answer: B) Large-scale image dataset

5. What does ?reduced cost of input? mean in platforms?

A) Higher expenses
B) Users provide content for free
C) More employees are needed
D) Less revenue

Answer: B) Users provide content for free

6. Which of the following is a revenue stream for platforms?

A) Deleting users
B) Virtual gifts
C) Reducing features
D) Free services only

Answer: B) Virtual gifts

7. What is a unicorn startup?

A) A startup with no funding
B) A public company
C) A private company valued over $1 billion
D) A small company

Answer: C) Private company valued over $1B

---

8. What are Porter?s two major competitive advantage types?

A) Growth and profit
B) Cost and differentiation
C) Sales and marketing
D) Technology and HR

Answer: B) Cost and Differentiation

---

9. What does the freemium model mean?

A) Everything is paid
B) Everything is free
C) Basic features are free and premium features are paid
D) Only ads are used

Answer: C) Basic free, premium paid

---

10. On what does collaborative filtering depend?

A) A single user only
B) Similar users? behavior
C) Hardware only
D) Random data

Answer: B) Similar users? behavior

Week 5

Concepts

1. Analytics as a Process

Analytics is not just about tools. It is a structured process for turning a business problem into a useful answer. If the process is weak, the result will also be weak.

Mindfulness means questioning the data, the assumptions, and the output. An analyst should not blindly trust software or numbers.

Parsimony, also called Occam?s Razor, means using the simplest model that works well. Too many variables can create noise, confusion, and overfitting.

2. Six-Step Analytics Lifecycle

The usual lifecycle is: define the problem, select variables, collect data, build the model, test the model, and implement and improve the solution.

The thought process is the most important step because it defines the business question, the output, and the approach.

3. Variable Selection and Data Quality

Variables can be chosen from prior research, domain experts, business usefulness, control needs, simplicity, and data availability.

Multicollinearity happens when variables are highly related to each other, such as age and date of birth. This can confuse the model.

A rough warning sign is correlation above 0.7. Another sign is a high VIF, where a value above 2 is considered problematic in these notes. The formula is VIF = 1 / (1 - R†).

Data quality includes accuracy, completeness, consistency, timeliness, believability, and interpretability.

Data should also be inspected manually. A strange value like zero sales may be a real event or a system error, so it should be checked before modeling.

4. Models, Learning, and Fit

A model is a simplified representation of reality. It tries to explain or predict an outcome using input variables.

Supervised learning uses labeled output variables. Unsupervised learning does not have labeled output and instead discovers patterns such as clusters.

RSS or Residual Sum of Squares measures the difference between actual and predicted values. Lower RSS means a better fit.

R† shows how much variation the model explains. It lies between 0 and 1. Higher values usually mean a better fit.

Standardization using Z-scores converts variables to a common scale with mean 0 and standard deviation 1. This helps especially in explanatory models.

Explanatory models focus on understanding why something happens. Predictive models focus on making accurate predictions, even if interpretation is less important.

5. Bias, Variance, and Validation

Bias comes from overly simple models and leads to underfitting. Variance comes from overly complex models that react too strongly to noise and leads to overfitting.

The bias-variance tradeoff means good modeling requires balance between simplicity and flexibility.

Training error is the error on data used to build the model. Test error is the error on unseen data and is more important for real performance.

Cross-validation helps estimate performance more reliably. In K-fold cross-validation, data is split into K parts and tested repeatedly. In LOOCV, K equals the number of observations.

6. Business Examples

The XYZ Bank case illustrates profitability skew, where a small percentage of customers contribute most of the profit.

Dummy regression uses 0 and 1 variables to represent categories and compare groups.

Correlation means variables move together. Causation means one variable actually causes the other. Confusing these two is a major analytical mistake.

> *Example: Ice cream sales and drowning cases may both rise in summer. They are correlated, but ice cream does not cause drowning. The real driver is hot weather.*

MCQs

1. What does process quality mainly determine in analytics?

A) Data size only
B) Final output quality
C) Model type only
D) Algorithm name

Answer: B

---

2. What does mindfulness mean in analytics?

A) Using more tools
B) Questioning data and results
C) Coding quickly
D) Storing data

Answer: B

---

3. What does parsimony mean?

A) Using a complex model
B) Using many variables
C) Using a simple model
D) Using no variables

Answer: C

---

4. What is the first step in the analytics lifecycle?

A) Testing
B) Thought process
C) Implementation
D) Data collection

Answer: B

---

5. What does the thought process step help define?

A) Coding style
B) The business problem
C) Data storage only
D) Model training only

Answer: B

---

6. What does a literature review mainly provide?

A) Raw data only
B) Scientific base and prior knowledge
C) Errors
D) Output labels only

Answer: B

---

7. What are control variables?

A) Variables that can be ignored
B) External factors that should be considered
C) Random numbers
D) Variables that must be removed

Answer: B

---

8. What should an analyst do if no data is available for a useful variable?

A) Ignore the issue completely
B) Use a proxy if possible
C) Guess the values
D) Stop all analysis

Answer: B

---

9. What does multicollinearity mean?

A) No relationship between variables
B) High relationship between variables
C) Random variables only
D) Missing data only

Answer: B

---

10. A correlation above which value is treated as a warning sign here?

A) 0.2
B) 0.5
C) 0.7
D) 1.0

Answer: C

---

11. A VIF above which value is treated as a problem here?

A) 1
B) 2
C) 5
D) 10

Answer: B

---

12. What is the formula for VIF?

A) R†
B) 1 - R†
C) 1 / (1 - R†)
D) R† - 1

Answer: C

---

13. What does data quality mainly affect?

A) Speed only
B) Accuracy of conclusions
C) UI design
D) Coding style

Answer: B

---

14. In the class example, what explained the zero-sales value?

A) A strike
B) A crisis
C) A system error
D) A marketing event

Answer: C

---

15. What is a model?

A) Exact reality
B) A simplified version of reality
C) Data only
D) Code only

Answer: B

---

16. What does supervised learning require?

A) No data
B) An output variable
C) Random input only
D) No model

Answer: B

---

17. Unsupervised learning is commonly used for what?

A) Prediction with labels
B) Classification only
C) Clustering
D) Regression only

Answer: C

---

18. What does RSS measure?

A) Fit error
B) Speed
C) Memory use
D) Data size

Answer: A

---

19. What does a lower RSS mean?

A) A worse model
B) A better fit
C) No model
D) Random behavior

Answer: B

---

20. What is the range of R†?

A) -1 to 1
B) 0 to 1
C) 1 to 10
D) 0 to 10

Answer: B

---

21. After Z-score standardization, what is the mean?

A) 1
B) 0
C) 10
D) -1

Answer: B

---

22. After Z-score standardization, what is the standard deviation?

A) 0
B) 1
C) 2
D) 10

Answer: B

---

23. What do explanatory models mainly focus on?

A) Prediction only
B) Accuracy only
C) Understanding why
D) Speed only

Answer: C

---

24. What do predictive models mainly focus on?

A) Why something happens
B) Accuracy
C) Explanation only
D) Variable count only

Answer: B

---

25. What does bias mean in modeling?

A) A complex-model problem
B) Error from an overly simple model
C) A perfect model
D) A data storage issue

Answer: B

---

26. What does variance mean in modeling?

A) Stability only
B) Sensitivity to noise
C) Accuracy only
D) Speed only

Answer: B

---

27. Overfitting is usually associated with which problem?

A) High bias
B) High variance
C) Low variance
D) No error

Answer: B

---

28. What is training error?

A) Real-world error only
B) Test-set error
C) Error on training data
D) Random error only

Answer: C

---

29. What is test error?

A) Error on unseen data
B) Error on training data
C) Coding error
D) Data entry error

Answer: A

---

30. Which metric matters more for real-world model performance?

A) Training error
B) Test error
C) Data size only
D) Speed only

Answer: B

---

31. What does cross-validation help with?

A) Speed only
B) Estimating model performance
C) Coding style
D) Storage design

Answer: B

---

32. In K-fold cross-validation, the data is split into how many parts?

A) 1
B) 2
C) K parts
D) N+1 parts

Answer: C

---

33. What does LOOCV mean?

A) K = 1
B) K = n
C) K = 0
D) K = 2

Answer: B

---

34. What does profitability skew mean?

A) Profit is equal across customers.
B) A few customers generate most of the profit.
C) There is no profit.
D) Only losses occur.

Answer: B

---

35. Which values are used in a dummy variable?

A) 1 and 2
B) 0 and 1
C) 2 and 3
D) -1 and 1

Answer: B

---

36. What does correlation mean?

A) Cause
B) Relationship
C) No relationship
D) Randomness only

Answer: B

---

37. What does causation mean?

A) Relationship only
B) Cause-and-effect
C) Pattern only
D) Data only

Answer: B

---

38. What is a common analytical mistake?

A) Ignoring data cleaning
B) Treating correlation as causation
C) Using models
D) Testing models

Answer: B

---

39. Which option is an example of a consistency issue in data quality?

A) Missing data
B) Unit mismatch
C) Outliers only
D) Noise only

Answer: B

---

40. What does believability mean in data quality?

A) The data looks good only.
B) The data makes sense and can be trusted.
C) The data is fast to process.
D) The data is large.

Answer: B

---

41. What does interpretability mean in data quality?

A) Data size
B) Ease of understanding the data
C) Coding style
D) Model speed

Answer: B

---

42. What does timeliness mean in data quality?

A) Old data is always better.
B) Data is available at the correct time.
C) Data is always big.
D) Data is random.

Answer: B

---

43. What is irreducible error?

A) An error that can always be fixed
B) Random noise that cannot be fully removed
C) A model choice problem only
D) A storage issue only

Answer: B

---

44. What usually defines the best model?

A) It is the most complex model.
B) It balances bias and variance well.
C) It uses the smallest dataset.
D) It uses no data.

Answer: B

---

45. What does overfitting often cause on new data?

A) High test error
B) Low test error
C) No error
D) Perfect prediction

Answer: A

---

46. What does underfitting usually cause?

A) High bias
B) Low bias
C) High variance
D) No variance

Answer: A

---

47. Standardization is especially useful for which type of model?

A) Prediction only
B) Explanation and comparison
C) Storage only
D) Cleaning only

Answer: B

---

48. What does adjusted R† do?

A) It is always higher than R†.
B) It penalizes unnecessary extra variables.
C) It removes data points.
D) It is always the same as R†.

Answer: B

---

49. Which quality best describes a good analyst?

A) Codes fast only
B) Questions data and assumptions
C) Uses many tools only
D) Copies a model

Answer: B

---

50. What is the final goal of analytics?

A) Coding only
B) Insight and decision support
C) Data storage only
D) UI design

Answer: B

Online Assessment Questions

1. Which source of business value from AI focuses on reducing costs by replacing human labor?

A) Coordination
B) Anticipation
C) Personalization
D) Automation

Answer: D) Automation

---

2. Through what is anticipation in AI mainly achieved?

A) Automating tasks
B) Reducing coordination costs
C) Predictive models and forecasting
D) Personalizing the user interface

Answer: C) Predictive models and forecasting

---

3. What does a business model mainly focus on?

A) External competition
B) Demand side only
C) Supply side: creating, delivering, and monetizing value
D) SWOT analysis

Answer: C) Supply side (create, deliver, monetize value)

---

4. In RBV theory, what must a resource be?

A) Easy to copy
B) Cheap
C) Valuable, rare, and hard to imitate
D) Only technology-based

Answer: C) Valuable, rare, and hard to imitate

---

5. Which of the following are complementary factors for AI capability?

A) People and talent
B) Market knowledge
C) Organizational culture
D) Training and incentives

Answer: A, B, C, D

---

6. What does Transaction Cost Economics compare?

A) Fixed cost and variable cost
B) Production cost and transaction cost
C) Profit and loss
D) Cost and revenue

Answer: B) Production + transaction cost

---

7. What is included in transaction cost?

A) Hardware cost
B) Training cost only
C) Coordination such as searching, contracting, and monitoring
D) Marketing cost

Answer: C) Coordination (searching, contracting, monitoring)

---

8. What are the two main components of transaction cost?

A) Fixed and variable cost
B) Coordination cost and transaction risk
C) Revenue and profit
D) Input and output

Answer: B) Coordination cost and transaction risk

---

9. What do economies of scale mean?

A) Cost increases with production
B) Cost decreases with higher production
C) Cost does not change
D) It applies only to services

Answer: B) Cost decreases with higher production

---

10. Which of the following are good KPIs for AI customer service?

A) Escalation rate
B) Average resolution time
C) Sentiment score
D) Total fixed cost

Answer: A, B, C

Week 6

Concepts

1. Classification and Targeting

Classification is the process of assigning observations to predefined classes, such as spam or not spam, likely buyer or unlikely buyer, churn or no churn. Because the correct class labels are known in training data, classification is a form of supervised learning.

Predefined classes are what make classification different from clustering. In classification, the categories already exist. In clustering, the groups are discovered from data.

A targeting problem is a common business use of classification. The goal is to identify the best subgroup, such as customers most likely to respond to a campaign.

Lift shows how much better a model performs compared with random selection. A lift greater than 1 means the model is useful.

2. Classification Algorithms

Logistic regression produces probabilities between 0 and 1. It is widely used for binary classification.

Naive Bayes is a probability-based classifier built on Bayes? theorem. It is simple and fast.

Discriminant analysis tries to find the boundary that separates classes well.

Decision trees split data step by step using rules, which makes them easy to understand.

AI methods such as Artificial Neural Networks (ANN) and Support Vector Machines (SVM) can capture complex patterns.

Clustering is normally unsupervised, but once clusters are labeled, they may sometimes be used in a classification-like way.

3. Model Evaluation and Confusion Matrix

Model fit shows how well the model explains the training data, but this alone is not enough.

Prediction performance shows how well the model works on new data, which matters more in practice.

A confusion matrix summarizes predictions using four outcomes: True Positive, True Negative, False Positive, and False Negative.

Accuracy is the proportion of all correct predictions. Sensitivity or recall measures how well the model finds actual positives. Specificity measures how well it identifies actual negatives. Precision measures how correct the positive predictions are. F1 score balances precision and recall.

4. ROC, PR, Gains, and Lift

A gains chart shows how effectively the model captures positives when observations are ranked.

A ROC curve plots True Positive Rate against False Positive Rate. A better model stays closer to the top-left corner.

AUC is the area under the ROC curve. A value of 0.5 means random performance, while 1.0 means perfect performance.

A PR curve plots precision against recall and is especially useful for imbalanced datasets.

5. Class Imbalance, Sampling, and Thresholds

Class imbalance happens when one class is much smaller than the other, such as fraud vs non-fraud. In such cases, accuracy can be misleading.

Over-sampling increases minority class representation. Under-sampling reduces the majority class. SMOTE creates synthetic minority examples.

The threshold converts predicted probabilities into class labels. A lower threshold gives more positive predictions and may increase false positives. A higher threshold gives fewer positive predictions and may increase false negatives.

6. Business Interpretation

The best metric depends on the business problem. In disease detection, missing a true case is dangerous, so sensitivity matters. In loan approval or fraud screening, false positives may be expensive, so specificity matters. In search or recommendation, precision may matter more.

> *Example: If only 1 out of 100 customers buys a product, a model that predicts ?no? for everyone gets 99% accuracy, but it is still useless. This is why accuracy alone is not enough for imbalanced data.*

MCQs

1. What is classification?

A) Grouping without labels
B) Assigning predefined classes
C) Regression only
D) Clustering only

Answer: B

---

2. Classification belongs to which type of learning?

A) Unsupervised learning
B) Supervised learning
C) Reinforcement learning
D) No learning type

Answer: B

---

3. Which algorithm produces probabilities between 0 and 1?

A) Decision tree only
B) Logistic regression
C) K-means
D) Clustering

Answer: B

---

4. What does a lift greater than 1 indicate?

A) A bad model
B) Performance equal to random
C) Better-than-random performance
D) A model error

Answer: C

---

5. What does a false negative mean?

A) A wrong positive prediction
B) A missed positive case
C) A correct negative case
D) A correct positive case

Answer: B

---

6. What is the formula for accuracy?

A) TP / Total
B) TN / Total
C) (TP + TN) / Total
D) TP + FP

Answer: C

---

7. What does sensitivity mainly focus on?

A) Negatives
B) Positives
C) Errors only
D) Accuracy only

Answer: B

---

8. What does specificity mainly focus on?

A) Positives
B) Negatives
C) Errors only
D) Recall only

Answer: B

---

9. What does precision measure?

A) Total correct predictions
B) Correctness of positive predictions
C) Total errors only
D) Recall only

Answer: B

---

10. What is the F1 score?

A) An average of all metrics
B) The harmonic mean of precision and recall
C) A simple sum of two metrics
D) A difference between two metrics

Answer: B

---

11. Clustering is mainly which type of learning?

A) Supervised
B) Unsupervised
C) Semi-supervised only
D) Reinforcement

Answer: B

---

12. What does a ROC curve show?

A) Accuracy versus loss
B) True Positive Rate versus False Positive Rate
C) Precision versus recall
D) Gain versus lift

Answer: B

---

13. What does an AUC of 0.5 mean?

A) Perfect model
B) Random model
C) Bad data only
D) High lift

Answer: B

---

14. For which type of data is the PR curve especially useful?

A) Balanced data only
B) Imbalanced data
C) Regression data only
D) Clustering data only

Answer: B

---

15. What does class imbalance mean?

A) Classes are equal in size
B) Classes are unequal in size
C) Data is missing
D) Data has errors

Answer: B

---

16. What does over-sampling do?

A) It removes data.
B) It increases the minority class.
C) It reduces features.
D) It increases errors.

Answer: B

---

17. What does under-sampling do?

A) It adds more data.
B) It reduces the majority class.
C) It increases features.
D) It does nothing.

Answer: B

---

18. What does SMOTE do?

A) It deletes data.
B) It creates synthetic minority data.
C) It guesses randomly.
D) It only sorts data.

Answer: B

---

19. Which statement best describes decision trees?

A) They are purely statistical equations only.
B) They are rule-based models.
C) They are random grouping methods.
D) They are unsupervised only.

Answer: B

---

20. What does Naive Bayes mainly use?

A) Trees
B) Probability
C) Neural networks
D) Sorting only

Answer: B

---

21. Which four terms appear in a confusion matrix?

A) TP, TN, FP, FN
B) TP only
C) Errors only
D) TN only

Answer: A

---

22. A false positive is also called which type of error?

A) Type I error
B) Type II error
C) Correct prediction
D) No error

Answer: A

---

23. A false negative is also called which type of error?

A) Type I error
B) Type II error
C) Correct prediction
D) No error

Answer: B

---

24. Which metric is especially important in cancer detection?

A) Accuracy
B) Precision
C) Sensitivity
D) Specificity

Answer: C

---

25. Which metric is especially important in loan approval or risk screening?

A) Recall only
B) Specificity
C) Accuracy only
D) Loss

Answer: B

---

26. What does a gains chart show?

A) Profit only
B) Model performance in capturing positives
C) Loss only
D) Error messages only

Answer: B

---

27. Lift compares model performance against what?

A) Random selection
B) Error rate only
C) Raw data only
D) Labels only

Answer: A

---

28. What usually happens if the threshold is set high?

A) More positives are predicted.
B) Fewer positives are predicted.
C) Accuracy becomes meaningless immediately.
D) Nothing changes.

Answer: B

---

29. What usually happens if the threshold is set low?

A) Fewer positives are predicted.
B) More positives are predicted.
C) Nothing changes.
D) The model stops working.

Answer: B

---

30. What does model fit refer to?

A) Performance on real future data only
B) Performance on training data
C) Performance on external labels only
D) No specific performance measure

Answer: B

---

31. Prediction performance is usually measured on which data?

A) Training data
B) Test data
C) Labels only
D) None of these

Answer: B

---

32. What is a targeting problem?

A) Random selection
B) Selecting the best subgroup
C) Sorting records only
D) No business use

Answer: B

---

33. What does ANN stand for?

A) Artificial Neural Network
B) Automated Node Network
C) Analytical Neural Node
D) None of these

Answer: A

---

34. What is SVM?

A) A tree only
B) A machine learning method
C) A clustering result only
D) None of these

Answer: B

---

35. Where should the best ROC curve lie?

A) Near a straight diagonal line
B) Near the top-left corner
C) Near the bottom line
D) At random points only

Answer: B

---

36. Accuracy becomes misleading mainly in which situation?

A) Balanced data
B) Imbalanced data
C) Clean data only
D) Small data only

Answer: B

---

37. If precision is high, what is usually true?

A) There are few false positives.
B) There are many false positives.
C) There are few true positives.
D) Nothing can be inferred.

Answer: A

---

38. If recall is high, what is usually true?

A) There are few false negatives.
B) There are many false negatives.
C) There are few false positives.
D) Nothing can be inferred.

Answer: A

---

39. When is the F1 score especially useful?

A) When only accuracy matters
B) When a balance between precision and recall is needed
C) When no data is available
D) It is never useful

Answer: B

---

40. What is the lift of a random model?

A) 0
B) 1
C) 2
D) 10

Answer: B

---

41. What does classification predict?

A) Numeric values only
B) Categories
C) Graphs only
D) Nothing

Answer: B

---

42. What does regression usually predict?

A) Categories
B) Numbers
C) Class labels only
D) Nothing

Answer: B

---

43. On what basis does clustering group data?

A) Known labels
B) Similarity
C) Hard-coded rules only
D) None

Answer: B

---

44. What is on the X-axis of the ROC curve?

A) True Positive Rate
B) False Positive Rate
C) Precision
D) Recall

Answer: B

---

45. What is on the Y-axis of the ROC curve?

A) Precision
B) Recall only
C) True Positive Rate
D) False Positive Rate

Answer: C

---

46. What is the formula for precision?

A) TP / (TP + FP)
B) TP / (TP + FN)
C) TN / (TN + FP)
D) None of these

Answer: A

---

47. What is the formula for recall?

A) TP / (TP + FN)
B) TP / (TP + FP)
C) TN / (TN + FP)
D) None of these

Answer: A

---

48. What is the formula for specificity?

A) TN / (TN + FP)
B) TP / (TP + FN)
C) TP / (TP + FP)
D) None of these

Answer: A

---

49. Which ratio is an example of highly imbalanced data?

A) 50:50
B) 99:1
C) 40:60
D) Equal classes

Answer: B

---

50. On what does the best evaluation metric mainly depend?

A) Data size only
B) The business case
C) The model name only
D) Random choice

Answer: B

Online Assessment Questions

1. Which metrics matter most to CEOs and CFOs for AI success?

A) Accuracy and precision
B) Speed of model training
C) Financial metrics such as revenue, cost savings, and CLV
D) Code efficiency

Answer: C) Financial metrics (revenue, cost savings, CLV)

---

2. What is the best method to establish causality in AI experiments?

A) Before-after analysis
B) Cohort analysis
C) A/B testing
D) Guessing

Answer: C) A/B testing

---

3. How does the Synthetic Control Method help?

A) By removing the need for a control group
B) By creating a comparable synthetic control
C) By reducing data
D) By only measuring revenue

Answer: B) Creating a comparable synthetic control

---

4. What is the biggest challenge in scaling AI projects?

A) Hardware cost
B) Data size
C) Proving ROI to stakeholders
D) Coding difficulty

Answer: C) Proving ROI to stakeholders

---

5. Why is accuracy misleading in imbalanced data?

A) The model is too complex.
B) Accuracy is always low.
C) The majority class dominates the accuracy value.
D) The wrong formula is used.

Answer: C) Majority class dominates accuracy

---

6. Which of the following are technical performance metrics?

A) AUC
B) Precision and Recall
C) KS Statistic
D) CLV

Answer: A, B, C

---

7. Which of the following are operational metrics?

A) Revenue
B) Process efficiency
C) Speed of delivery
D) Automation

Answer: B, C, D

---

8. Which of the following are ranking metrics?

A) Mean Reciprocal Rank (MRR)
B) NDCG
C) Precision
D) F1-score

Answer: A, B

---

9. What does Swap Set Analysis measure?

A) Accuracy
B) Profit under constraints
C) Speed
D) Data size

Answer: B) Profit under constraints

---

10. What does the KS Statistic measure?

A) Accuracy
B) Loss
C) Separation between classes
D) Training time

Answer: C) Separation between classes